HW #5: Code

Submit your solution on Canvas (see end for submission checklist).

For the following questions, consider the Stanford Manipulator shown in Figure 1.

The local frames of each link are show as $x_0 y_0 z_0$ for the base link to $x_6 y_6 z_6$ for end-effector link 6. The joint axis are about the z axis of the previous link, such that $\boldsymbol{\omega}_1 = z_0$, $\boldsymbol{\omega}_2 = z_1$, etc. Each axis is a revolute joint, except for $\boldsymbol{\omega}_3$, which is prismatic.

For the distances d_{1-6} use the following:

$d_1 = 40$ cm
$d_2 = 50$ cm
$d_3 = 60$ cm
$d_6 = 30$ cm

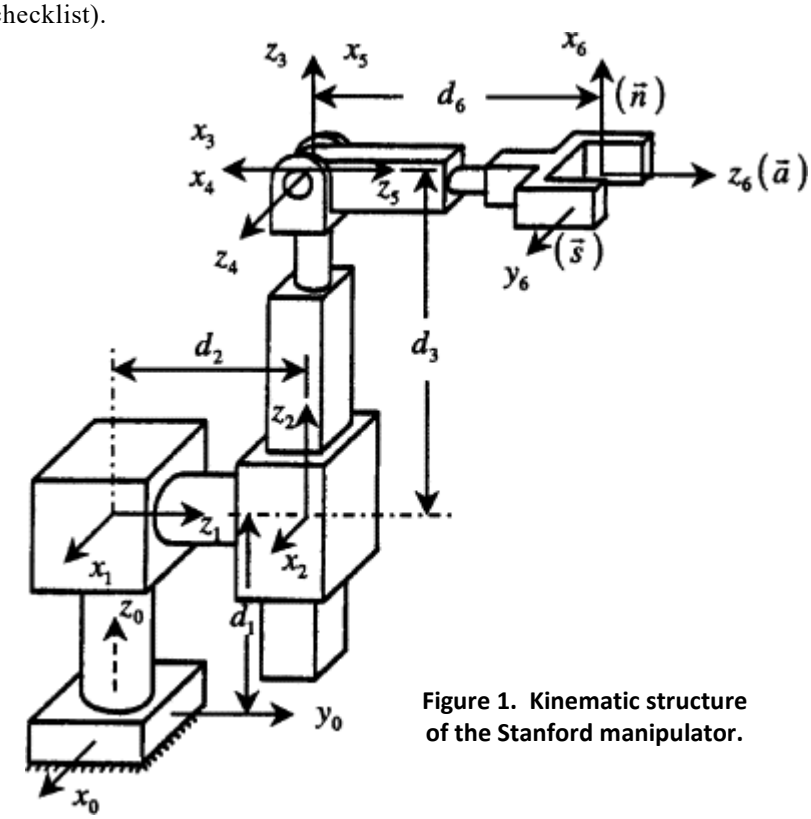

Figure 1. Kinematic structure of the Stanford manipulator.

1) Using MATLAB, find the forward kinematics, $\mathbf{g}_{st}(\theta)$, of the Stanford manipulator above. Use the posted sub-functions. Name your joint angles th1, th2, etc. Use the simplify command to simplify the matrix. Your script should output only gstth as a symbolic object. It should be a function of only the joint angles.
2) Use the matlabFunction command to create a function called gstthCKB.m (use your own initials). The function should take all 6 angles as a vector, rather than individual angles.

HW #5: WHAT TO SUBMIT

Submit ***individual files*** on Canvas (not zipped) using the naming convention given. Check Canvas for the due date and time.

A. **HW05_Lastname.m /.py:** Your kinematics m-file. This file should output the symbolic object **gstth** and create the function **gstthCKB.m/.py** (use your initials, not mine!).

Solution Hints:

```
>> gstthCKB([0 0 0 0 0 0])

ans =

    0    1    0    0
    0    0    1   80
    1    0    0  100
    0    0    0    1
```

```
>> gstthCKB([30 60 0 -60 60 30]*pi/180)

ans =

    0.5558    0.4291    0.7120   41.3606
   -0.6121   -0.3683    0.6998   90.2748
    0.5625   -0.8248    0.0580   71.7404
         0         0         0    1.0000
```